

WILLIAM LEE

Senior Software Engineer - Full Stack, Cloud, Data and AI

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PROFESSIONAL SUMMARY

Senior Software Engineer with 11+ years delivering secure products across security, device services, learning technology, and digital media, specializing in React 18-19, Node.js 18-22, Python 3.10-3.12, Kubernetes, and cloud data services on AWS and Azure; combines architecture, production debugging, modernization, observability, and leadership to build scalable systems that improve reliability, delivery speed, user experience, and measurable business outcomes.

SKILLS

Languages & Frameworks: JavaScript/TypeScript, React, Node.js, Java/Spring Boot, C#/.NET, Python/FastAPI, Go, SQL, HTML5, CSS3

Cloud & DevOps: AWS, Azure, GCP, Docker, Kubernetes 1.24-1.31, Terraform 1.x, GitHub Actions, Jenkins, serverless, Linux, networking, IAM, cost optimization

Data & AI: PostgreSQL, MySQL, SQL Server, MongoDB, Redis, Kafka, Spark, Airflow, data pipelines, vector databases, RAG, LLM integration, model evaluation

Architecture & APIs: REST, GraphQL, gRPC, WebSockets, microservices, event-driven systems, modular monoliths,DDD, CQRS, API versioning, idempotency, resilience patterns

Quality & Observability: Jest, JUnit, pytest, xUnit, Cypress, Playwright, Testcontainers, OpenTelemetry, Prometheus, Grafana, Datadog, CloudWatch, ELK/OpenSearch

Security & Leadership: OAuth 2.0, OIDC, RBAC, SAST, SCA, threat modeling, secure SDLC, architecture reviews, mentoring, incident response, Agile delivery, production ownership

WORK EXPERIENCE

Self Employed | Tennessee, United States

Staff Software Engineer | June 2026 - Present

- Architected and delivered client-facing products with **React, Node.js, Java, Python, cloud services, and data pipelines** on **AWS and Azure**, translating ambiguous product goals into secure roadmaps, technical designs, and production-ready increments.
- Modernized legacy applications toward **React 18-19, Node.js 18-22, Python 3.10-3.12, Kubernetes, and cloud data services**, replacing tightly coupled modules with **modular services, event-driven integration, and cloud-native delivery** while preserving business behavior through characterization and regression tests.
- Diagnosed difficult production issues involving **memory leaks, slow queries, deployment regressions, and inconsistent API behavior**, using distributed traces, structured logs, browser profiling, and database plans to isolate root causes before release.
- Implemented new subscription, reporting, workflow, and administrative features across **full-stack platforms, APIs, automation, and analytics**, balancing delivery speed with backward compatibility, accessibility, and operational support.
- Introduced **API versioning, schema validation, feature flags, and automated rollback controls**, reducing release risk and allowing incremental migration of customers from older application paths.
- Built **CI/CD pipelines, infrastructure automation, and quality gates** with GitHub Actions, containers, and cloud-native deployment services, making releases repeatable across development and production.
- Strengthened **authentication, authorization, secrets management, dependency scanning, and audit logging**, addressing insecure defaults and closing gaps discovered during threat modeling and code review.
- Mentored engineers through architecture reviews, pairing, incident retrospectives, and pragmatic coding standards, improving maintainability without blocking teams with unnecessary process overhead.
- Improved median response and interaction performance by **31%** through caching, query tuning, bundle reduction, and targeted concurrency changes, increasing reliability during peak customer activity.

Veracode | Somerville, MA

Staff Software Engineer | July 2021 - May 2026

- Led development of application-security platform capabilities using **React, Node.js, Java, Python, cloud services, and data pipelines** with **AWS and Azure**, supporting multi-tenant workflows for scan submission, policy evaluation, findings review, and remediation tracking.
- Upgraded aging modules from period-appropriate foundations to modern supported releases, resolving deprecated APIs, brittle build scripts, and inconsistent behavior without disrupting enterprise customers.
- Resolved production defects where high-volume scan jobs caused **timeouts, duplicate processing, stale status displays, and database contention**, adding idempotency, backpressure, and clearer operational telemetry.
- Designed new policy, notification, search, and role-management features with **contract-first APIs and backward-compatible schemas**, enabling product teams to ship functionality without breaking integrations.
- Refactored shared platform components into **well-bounded services and reusable modules**, lowering coupling between scan orchestration, policy calculation, identity, reporting, and customer configuration.
- Expanded automated coverage with **unit, integration, contract, end-to-end, security, and performance tests**, catching release regressions that previously appeared only in large customer environments.
- Partnered with product, security research, support, and SRE teams to triage critical incidents, communicate impact, validate fixes, and convert recurring failure patterns into backlog improvements.
- Reduced customer-visible scan workflow failures by **38%** after improving retry policy, queue observability, validation, and failure recovery, while maintaining secure handling of sensitive analysis data.

Asurion | Nashville, TN

Senior Software Engineer | May 2018 - June 2021

- Developed customer and agent experiences with **React, Node.js, Java, Python, cloud services, and data pipelines** for device protection, enrollment, claims, replacement logistics, and support workflows operating across web, mobile, and service channels.

- Modernized period-appropriate legacy components and services, separating presentation, business rules, and integrations so teams could release features without rebuilding the entire claims application.
- Investigated incidents involving **payment retries, inventory mismatches, slow eligibility checks, session loss, and third-party carrier errors**, adding resilient fallbacks and actionable monitoring.
- Implemented new claims status, appointment, notification, and self-service features using **secure APIs, asynchronous events, and tested user journeys**, improving transparency for customers and support teams.
- Improved data consistency through **idempotent commands, optimistic concurrency, reconciliation jobs, and schema constraints**, preventing duplicate replacements and incorrect workflow transitions.
- Collaborated with QA, operations, product, and vendor teams during staged releases, using feature toggles and production validation to limit risk across high-traffic customer journeys.
- Cut average claim-processing latency by **27%** through query optimization, response caching, parallel integration calls, and removal of redundant client requests during peak device-launch periods.

Rustici Software | Nashville, TN

Software Engineer | February 2016 - March 2018

- Built learning-technology capabilities around **SCORM, xAPI, course launch, learner state, and reporting**, using period-appropriate web and service technologies with careful interoperability testing.
- Debugged cross-browser launch failures, malformed content packages, inconsistent completion rules, and customer-specific integration errors by tracing communication across players, APIs, and learning systems.
- Developed new administration, analytics, import, and content-validation features while maintaining compatibility with older browsers, established learning standards, and customer-hosted environments.
- Improved maintainability by extracting shared parsing, validation, and persistence logic into tested modules, reducing duplicated behavior across content launch and reporting workflows.
- Supported releases with automated tests, deployment checklists, customer issue reproduction, and post-release monitoring, helping the team resolve production defects with clear technical evidence.
- Collaborated directly with product and customer-facing teams to explain standards behavior, estimate complex integration work, and shape practical fixes for ambiguous vendor implementations.

Mediacube | Knoxville, TN

Junior Software Engineer | June 2015 - January 2016

- Implemented early-career features for digital-media dashboards, creator reporting, metadata management, and operational tools using period-appropriate web technologies and relational databases.
- Fixed defects involving incorrect revenue totals, failed media imports, inconsistent date handling, and slow administrative searches by reproducing data conditions and adding targeted tests.
- Created reusable UI and service components for account management, reporting filters, exports, and content status workflows, improving consistency across rapidly evolving product areas.
- Assisted with database migrations, release verification, production support, and log analysis, gaining practical experience with safe deployment and customer-impact assessment.
- Participated in code reviews and Agile planning while documenting implementation decisions, known limitations, and support procedures for features delivered to internal and external users.

SELECTED PROJECTS

Secure Multi-Tenant Product Platform

Architected a cloud-native product spanning web applications, APIs, messaging, relational data, analytics, and AI-assisted workflows; introduced modular boundaries, automated testing, observability, feature flags, and progressive delivery to support reliable change across customer-facing and operational functions.

Legacy Modernization and Developer Platform

Created reusable infrastructure, CI/CD, service templates, design standards, and operational controls while incrementally replacing aging modules; reduced coupling, improved deployment consistency, and provided teams with documented paved roads for secure full-stack delivery.

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

Verified certification details were not provided; add only credentials personally earned before submission. Role-aligned professional development covers **AWS and Azure solution architecture, Kubernetes application development, secure software lifecycle, and role-specific framework training**, supported by practical architecture, delivery, production support, and security-focused engineering work.

EDUCATION

University of Tennessee, Knoxville | Bachelor of Science, Computer Science | September 2011 - May 2013